CYBERSRS

Secure Remote Access System

Software Requirements Specification

DRAFT — human review required

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AI-generated advisory document. Review by a qualified professional is required.

Automated checks do not prove completeness or semantic correctness. SDLC suitability scores are rule-based points, not probabilities.

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1. Introduction

Original project description

Design a secure university VPN and remote-access management system that allows students, faculty, researchers, and administrative staff to securely access approved campus resources from off-campus locations. The system should use centralized authentication, role-based access control, multi-factor authentication for privileged users, encrypted remote connections, session monitoring, and automatic security alerts. Sensitive administrative and research systems should only be accessible to specifically authorized users, while guest users must not receive VPN access. Network administrators should be able to create and revoke remote-access permissions, review active sessions, terminate suspicious sessions, manage VPN policies, and inspect audit logs. The system should also detect repeated failed login attempts, unusual login locations, excessive session duration, and unauthorized attempts to access restricted internal resources. All authentication events, access decisions, policy changes, and administrator actions must be logged. The solution should be secure, scalable, reliable, and support both university-managed and approved personal devices.

Summary: This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

Items

Stakeholder source statement E-001

Detail: Design a secure university VPN and remote-access management system that allows students, faculty, researchers, and administrative staff to securely access approved campus resources from off-campus locations.

Status: confirmed

Evidence ids: E-001

Stakeholder source statement E-002

Detail: The system should use centralized authentication, role-based access control, multi-factor authentication for privileged users, encrypted remote connections, session monitoring, and automatic security alerts.

Status: confirmed

Evidence ids: E-002

Out of scope: stakeholder exclusion

Detail: Sensitive administrative and research systems should only be accessible to specifically authorized users, while guest users must not receive VPN access.

Status: confirmed

Evidence ids: E-003

Stakeholder source statement E-004

Detail: Network administrators should be able to create and revoke remote-access permissions, review active sessions, terminate suspicious sessions, manage VPN policies, and inspect audit logs.

Status: confirmed

Evidence ids: E-004

Stakeholder source statement E-005

Detail: The system should also detect repeated failed login attempts, unusual login locations, excessive session duration, and unauthorized attempts to access restricted internal resources.

Status: confirmed

Evidence ids: E-005

Stakeholder source statement E-006

Detail: All authentication events, access decisions, policy changes, and administrator actions must be logged.

Status: confirmed

Evidence ids: E-006

Stakeholder source statement E-007

Detail: The solution should be secure, scalable, reliable, and support both university-managed and approved personal devices.

Status: confirmed

Evidence ids: E-007

2. Functional requirements

Applicability reason: Source-backed obligations; unspecified decisions remain open.

Requirements

FR-001 - allow students, faculty, researchers, and administrative staff to securely access approved campus resources from off-campus locations.

Statement: The system shall allow students, faculty, researchers, and administrative staff to securely access approved campus resources from off-campus locations.

Rationale: Implements stakeholder source E-001.

Priority: must

Dimensions: business

Basis: explicit

Evidence ids: E-001

Acceptance criteria

Given: An authenticated user attempting remote access

When: The user connects via the university VPN

Then: The connection is established with encrypted tunneling and validated credentials

Method: test

Observable evidence: Test capture of successful remote access session

Unresolved decisions: Which specific campus resources are approved for off-campus access

FR-002 - use centralized authentication, role-based access control, multi-factor authentication for privileged users, encrypted remote connections, session monitoring, and automatic security alerts.

Statement: The system shall use centralized authentication, role-based access control, multi-factor authentication for privileged users, encrypted remote connections, session monitoring, and automatic security alerts.

Rationale: Implements stakeholder source E-002.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: A user with elevated privileges attempting to access a restricted resource

When: The user performs an action requiring MFA verification

Then: The system prompts for multi-factor authentication and grants access only after successful completion

Method: demonstration

Observable evidence: Test capture of MFA prompt and successful access flow

Unresolved decisions: How role-based access is defined and updated

FR-003 - Sensitive administrative and research systems shall be accessible only to specifically authorized users, while guest users must not receive VPN access.

Statement: Sensitive administrative and research systems shall be accessible only to specifically authorized users, while guest users must not receive VPN access.

Rationale: Implements stakeholder source E-003.

Priority: must

Dimensions: business

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: A guest user attempting to connect via the university VPN

When: The system evaluates the user's authorization status

Then: The connection is denied with a clear message indicating unauthorized access

Method: test

Observable evidence: Test capture of denial response for guest user

Unresolved decisions: How 'specifically authorized' users are identified and maintained

FR-004 - Network administrators shall be able to create and revoke remote-access permissions, review active sessions, terminate suspicious sessions, manage VPN policies, and inspect audit logs.

Statement: Network administrators shall be able to create and revoke remote-access permissions, review active sessions, terminate suspicious sessions, manage VPN policies, and inspect audit logs.

Rationale: Implements stakeholder source E-004.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: An administrator with appropriate privileges

When: The administrator initiates a session termination for a suspicious connection

Then: The system terminates the session and updates the audit log with a timestamp and reason

Method: inspection

Observable evidence: Test capture of session termination event in audit log

Unresolved decisions: Which actions require administrative approval

FR-005 - detect repeated failed login attempts, unusual login locations, excessive session duration, and unauthorized attempts to access restricted internal resources.

Statement: The system shall detect repeated failed login attempts, unusual login locations, excessive session duration, and unauthorized attempts to access restricted internal resources.

Rationale: Implements stakeholder source E-005.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-005

Acceptance criteria

Given: A user attempting multiple failed logins from a new location

When: The system monitors the login pattern and triggers an alert

Then: An automated security alert is generated and sent to the appropriate team

Method: analysis

Observable evidence: Test capture of failed login sequence and subsequent alert trigger

Unresolved decisions: What constitutes 'excessive' session duration or 'unusual' location

3. Non-functional requirements

Applicability reason: Source-backed obligations; unspecified decisions remain open.

Requirements

NFR-001 - The solution shall be secure, scalable, reliable, and support both university-managed and approved personal devices.

Statement: The solution shall be secure, scalable, reliable, and support both university-managed and approved personal devices.

Rationale: Implements stakeholder source E-007.

Priority: should

Dimensions: availability

Basis: explicit

Evidence ids: E-007

Acceptance criteria

Given: A user with an approved personal device attempts to access campus resources from off-campus

When: The system evaluates device compliance and authentication status

Then: The system shall permit access only if the device is university-approved or managed, and authenticated securely

Method: test

Observable evidence: Test capture of device eligibility determination and access decision

Given: A user with a non-university-managed personal device attempts to connect via VPN

When: The system performs device validation against approved device policies

Then: The system shall deny access and return a policy violation message

Method: test

Observable evidence: Test capture of access denial due to unapproved device type

Given: A network administrator configures a new device policy for personal devices

When: The policy is applied to the system configuration

Then: The system shall enforce the updated policy during subsequent access attempts

Method: inspection

Observable evidence: Test capture of policy enforcement behavior on next access attempt

Unresolved decisions: Definition of 'approved personal devices'; Thresholds for device validation (e.g., OS version, encryption status); Mechanism for user-level device enrollment or certification

4. Security, privacy and compliance

Applicability reason: Source-backed obligations; unspecified decisions remain open.

Requirements

SEC-001 - All authentication events, access decisions, policy changes, and administrator actions must be logged.

Statement: All authentication events, access decisions, policy changes, and administrator actions must be logged.

Rationale: Implements stakeholder source E-006.

Priority: must

Dimensions: security

Basis: explicit

Evidence ids: E-006

Acceptance criteria

Given: An authenticated user performs a login or access decision

When: The system processes the action

Then: A log entry is created with timestamp, user ID, action type, and outcome

Method: test

Observable evidence: Test capture of authentication event logs

Given: An administrator modifies a policy or revokes access

When: The change is applied to the system

Then: A log entry is generated with timestamp, actor ID, policy name, and change details

Method: inspection

Observable evidence: Audit log record of policy modification

Given: A user attempts an action that results in failure or success

When: The action completes

Then: A log entry is created with outcome status and relevant context

Method: demonstration

Observable evidence: Test capture of access decision logs

Unresolved decisions: Which specific fields are required in the log entries (e.g., IP address, device type)?; What is the retention period for audit logs?; How will logs be accessed by administrators without exposing sensitive data?

Related requirement ids: FR-002; FR-004

5. Data requirements

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-004; SEC-001

6. Network and interfaces

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-003; FR-004

7. Compliance and policy applicability matrix

Open decisions: Which specific campus resources are approved for off-campus access; How role-based access is defined and updated; How 'specifically authorized' users are identified and maintained; Which actions require administrative approval; What constitutes 'excessive' session duration or 'unusual' location; Definition of 'approved personal devices','Thresholds for device validation (e.g., OS version, encryption status)','Mechanism for user-level device enrollment or certification; Which specific fields are required in the log entries (e.g., IP address, device type)?; What is the retention period for audit logs?; How will logs be accessed by administrators without exposing sensitive data?

8. Architecture

Summary: Proposed logical components and interactions mapped to canonical source obligations.

Items

Authentication Engine

Detail: Proposed logical owner for FR-002, SEC-001, FR-001.

Status: proposed

Evidence ids: E-001; E-002; E-006

Requirement ids: FR-002; SEC-001; FR-001

Component name: Authentication Engine

Responsibility: The system shall use centralized authentication, role-based access control, multi-factor authentication for privileged users, encrypted remote connections, session monitoring, and automatic security alerts. All authentication events, access decisions, policy changes, and administrator actions must be logged. The system shall allow students, faculty, researchers, and administrative staff to securely access approved campus resources from off-campus locations.

Access Control Module

Detail: Proposed logical owner for FR-003, FR-004, NFR-001.

Status: proposed

Evidence ids: E-003; E-004; E-007

Requirement ids: FR-003; FR-004; NFR-001

Component name: Access Control Module

Responsibility: Sensitive administrative and research systems shall be accessible only to specifically authorized users, while guest users must not receive VPN access. Network administrators shall be able to create and revoke remote-access permissions, review active sessions, terminate suspicious sessions, manage VPN policies, and inspect audit logs. The solution shall be secure, scalable, reliable, and support both university-managed and approved personal devices.

Session Monitoring and Alerting Service

Detail: Proposed logical owner for FR-005, SEC-001, FR-001.

Status: proposed

Evidence ids: E-001; E-005; E-006

Requirement ids: FR-005; SEC-001; FR-001

Component name: Session Monitoring and Alerting Service

Responsibility: The system shall detect repeated failed login attempts, unusual login locations, excessive session duration, and unauthorized attempts to access restricted internal resources. All authentication events, access decisions, policy changes, and administrator actions must be logged. The system shall allow students, faculty, researchers, and administrative staff to securely access approved campus resources from off-campus locations.

Network Policy Manager

Detail: Proposed logical owner for FR-004, FR-001.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-004; FR-001

Component name: Network Policy Manager

Responsibility: Network administrators shall be able to create and revoke remote-access permissions, review active sessions, terminate suspicious sessions, manage VPN policies, and inspect audit logs. The system shall allow students, faculty, researchers, and administrative staff to securely access approved campus resources from off-campus locations.

Data flows

Proposed source-defined interaction

Detail: Proposed directed path for FR-001, SEC-001.

Status: proposed

Evidence ids: E-001; E-006

Requirement ids: FR-001; SEC-001

From component: Authentication Engine

To component: Session Monitoring and Alerting Service

Data or action: The system shall allow students, faculty, researchers, and administrative staff to securely access approved campus resources from off-campus locations. All authentication events, access decisions, policy changes, and administrator actions must be logged.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: FR-001

From component: Authentication Engine

To component: Network Policy Manager

Data or action: The system shall allow students, faculty, researchers, and administrative staff to securely access approved campus resources from off-campus locations.

Proposed source-defined interaction

Detail: Proposed directed path for FR-004.

Status: proposed

Evidence ids: E-004

Requirement ids: FR-004

From component: Access Control Module

To component: Network Policy Manager

Data or action: Network administrators shall be able to create and revoke remote-access permissions, review active sessions, terminate suspicious sessions, manage VPN policies, and inspect audit logs.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001, SEC-001.

Status: proposed

Evidence ids: E-001; E-006

Requirement ids: FR-001; SEC-001

From component: Session Monitoring and Alerting Service

To component: Authentication Engine

Data or action: The system shall allow students, faculty, researchers, and administrative staff to securely access approved campus resources from off-campus locations. All authentication events, access decisions, policy changes, and administrator actions must be logged.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: FR-001

From component: Session Monitoring and Alerting Service

To component: Network Policy Manager

Data or action: The system shall allow students, faculty, researchers, and administrative staff to securely access approved campus resources from off-campus locations.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: FR-001

From component: Network Policy Manager

To component: Authentication Engine

Data or action: The system shall allow students, faculty, researchers, and administrative staff to securely access approved campus resources from off-campus locations.

Proposed source-defined interaction

Detail: Proposed directed path for FR-004.

Status: proposed

Evidence ids: E-004

Requirement ids: FR-004

From component: Network Policy Manager

To component: Access Control Module

Data or action: Network administrators shall be able to create and revoke remote-access permissions, review active sessions, terminate suspicious sessions, manage VPN policies, and inspect audit logs.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: FR-001

From component: Network Policy Manager

To component: Session Monitoring and Alerting Service

Data or action: The system shall allow students, faculty, researchers, and administrative staff to securely access approved campus resources from off-campus locations.

Trust boundaries

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-001, SEC-001.

Status: proposed

Evidence ids: E-001; E-006

Requirement ids: FR-001; SEC-001

Untrusted component: Authentication Engine

Trusted component: Session Monitoring and Alerting Service

Enforcement: The system shall allow students, faculty, researchers, and administrative staff to securely access approved campus resources from off-campus locations. All authentication events, access decisions, policy changes, and administrator actions must be logged.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: FR-001

Untrusted component: Authentication Engine

Trusted component: Network Policy Manager

Enforcement: The system shall allow students, faculty, researchers, and administrative staff to securely access approved campus resources from off-campus locations.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-004.

Status: proposed

Evidence ids: E-004

Requirement ids: FR-004

Untrusted component: Access Control Module

Trusted component: Network Policy Manager

Enforcement: Network administrators shall be able to create and revoke remote-access permissions, review active sessions, terminate suspicious sessions, manage VPN policies, and inspect audit logs.

9. Threats

Assumptions

1. These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

10. Risks

11. User stories and use cases

User stories and use cases

Id: UC-001

Actor: administrative staff

Goal: The system shall allow students, faculty, researchers, and administrative staff to securely access approved campus resources from off-campus locations.

Story: As administrative staff, I need the following behavior: The system shall allow students, faculty, researchers, and administrative staff to securely access approved campus resources from off-campus locations.

Preconditions

1. An authenticated user attempting remote access

Steps

1. The user connects via the university VPN

Success outcome: The connection is established with encrypted tunneling and validated credentials

Requirement ids: FR-001

Evidence ids: E-001

Unresolved decisions: Which specific campus resources are approved for off-campus access

Open issues: Which specific campus resources are approved for off-campus access

12. Verification and acceptance

Tests

Id: TEST-001

Requirement ids: FR-001

Procedure

1. Given: An authenticated user attempting remote access
2. When: The user connects via the university VPN

Expected result: Expected: The connection is established with encrypted tunneling and validated credentials

Evidence to record: Evidence: Test capture of successful remote access session

Test type: positive

Id: TEST-002

Requirement ids: FR-002

Procedure

1. Given: A user with elevated privileges attempting to access a restricted resource
2. When: The user performs an action requiring MFA verification

Expected result: Expected: The system prompts for multi-factor authentication and grants access only after successful completion

Evidence to record: Evidence: Test capture of MFA prompt and successful access flow

Test type: positive

Id: TEST-003

Requirement ids: FR-003

Procedure

1. Given: A guest user attempting to connect via the university VPN
2. When: The system evaluates the user's authorization status

Expected result: Expected: The connection is denied with a clear message indicating unauthorized access

Evidence to record: Evidence: Test capture of denial response for guest user

Test type: negative

Id: TEST-004

Requirement ids: FR-004

Procedure

1. Given: An administrator with appropriate privileges
2. When: The administrator initiates a session termination for a suspicious connection

Expected result: Expected: The system terminates the session and updates the audit log with a timestamp and reason

Evidence to record: Evidence: Test capture of session termination event in audit log

Test type: inspection

Id: TEST-005

Requirement ids: FR-005

Procedure

1. Given: A user attempting multiple failed logins from a new location
2. When: The system monitors the login pattern and triggers an alert

Expected result: Expected: An automated security alert is generated and sent to the appropriate team

Evidence to record: Evidence: Test capture of failed login sequence and subsequent alert trigger

Test type: analysis

Id: TEST-006

Requirement ids: NFR-001

Procedure

1. Given: A user with an approved personal device attempts to access campus resources from off-campus
2. When: The system evaluates device compliance and authentication status

Expected result: Expected: The system shall permit access only if the device is university-approved or managed, and authenticated securely

Evidence to record: Evidence: Test capture of device eligibility determination and access decision

Test type: positive

Id: TEST-007

Requirement ids: NFR-001

Procedure

1. Given: A user with a non-university-managed personal device attempts to connect via VPN
2. When: The system performs device validation against approved device policies

Expected result: Expected: The system shall deny access and return a policy violation message

Evidence to record: Evidence: Test capture of access denial due to unapproved device type

Test type: positive

Id: TEST-008

Requirement ids: NFR-001

Procedure

1. Given: A network administrator configures a new device policy for personal devices
2. When: The policy is applied to the system configuration

Expected result: Expected: The system shall enforce the updated policy during subsequent access attempts

Evidence to record: Evidence: Test capture of policy enforcement behavior on next access attempt

Test type: inspection

Id: TEST-009

Requirement ids: SEC-001

Procedure

1. Given: An authenticated user performs a login or access decision
2. When: The system processes the action

Expected result: Expected: A log entry is created with timestamp, user ID, action type, and outcome

Evidence to record: Evidence: Test capture of authentication event logs

Test type: positive

Id: TEST-010

Requirement ids: SEC-001

Procedure

1. Given: An administrator modifies a policy or revokes access

2. When: The change is applied to the system

Expected result: Expected: A log entry is generated with timestamp, actor ID, policy name, and change details

Evidence to record: Evidence: Audit log record of policy modification

Test type: inspection

Id: TEST-011

Requirement ids: SEC-001

Procedure

1. Given: A user attempts an action that results in failure or success
2. When: The action completes

Expected result: Expected: A log entry is created with outcome status and relevant context

Evidence to record: Evidence: Test capture of access decision logs

Test type: negative

Limitations: These verification plans have not been executed; their acceptance outcomes require testing.

13. Glossary

Summary: Source-present terminology; definitions introduce no additional product obligation.

Items

Policy

Detail: A rule or decision governing permitted actions; its project-specific meaning remains defined by the source requirements.

Status: proposed

Evidence ids: E-004; E-006

VPN

Detail: Virtual private network: a logical private network connection over another network.

Status: proposed

Evidence ids: E-001; E-003; E-004

Multi-factor authentication

Detail: Authentication using more than one distinct factor; this definition does not select products or protocols.

Status: proposed

Evidence ids: E-002

Access control

Detail: Rules and enforcement governing which principals may perform actions on resources.

Status: proposed

Evidence ids: E-002

Log

Detail: A collection of event records. This term does not itself require additional logging fields, events or access permissions.

Status: proposed

Evidence ids: E-004

Audit record

Detail: A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

Status: proposed

Evidence ids: E-004

14. SDLC recommendation and workflow

Selected model: Agile

Justification: No differentiating input evidence is provided to favor one SDLC model over others. All models receive equal suitability points (40) due to absence of factors such as budget, delivery, failure consequences, feedback, legacy, regulation, schedule, stability, uncertainty, verification. The decision remains provisional until these unknown factors are clarified.

Tradeoffs

1. Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.

Phases

Name: Requirements and verification planning

Phase type: planning

Activities

1. Review and baseline the source-defined work group: FR-001: The system shall allow students, faculty, researchers, and administrative staff to securely access approved campus resources from off-campus locations.; FR-002: The system shall use centralized authentication, role-based access control, multi-factor authentication for privileged users, encrypted remote connections, session monitoring, and automatic security alerts.; FR-003: Sensitive administrative and research systems shall be accessible only to specifically authorized users, while guest users must not receive VPN access.; FR-004: Network administrators shall be able to create and revoke remote-access permissions, review active sessions, terminate suspicious sessions, manage VPN policies, and inspect audit logs.; FR-005: The system shall detect repeated failed login attempts, unusual login locations, excessive session duration, and unauthorized attempts to access restricted internal resources.
2. Review and baseline the source-defined work group: NFR-001: The solution shall be secure, scalable, reliable, and support both university-managed and approved personal devices.; SEC-001: All authentication events, access decisions, policy changes, and administrator actions must be

logged.

3. Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Versioned requirements baseline
2. Unit, integration, system and acceptance test plan
3. Unresolved-decision record

Entry criteria

1. Authoritative inputs and canonical requirements are available for review.

Exit criteria

1. A human reviewer has recorded the baseline decision and disposition of unresolved scope.

Security checks

1. Review implementation and acceptance evidence against SEC-001: All authentication events, access decisions, policy changes, and administrator actions must be logged.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; NFR-001; SEC-001

Name: Agile delivery cycle

Phase type: iteration

Activities

1. Review and prioritize the backlog for the next iteration without changing source-defined scope.
2. Implement the source-defined work group: FR-001: The system shall allow students, faculty, researchers, and administrative staff to securely access approved campus resources from off-campus locations.; FR-002: The system shall use centralized authentication, role-based access control, multi-factor authentication for privileged users, encrypted remote connections, session monitoring, and automatic security alerts.; FR-003: Sensitive administrative and research systems shall be accessible only to specifically authorized users, while guest users must not receive VPN access.; FR-004: Network administrators shall be able to create and revoke remote-access permissions, review active sessions, terminate suspicious sessions, manage VPN policies, and inspect audit logs.; FR-005: The system shall detect repeated failed login attempts, unusual login locations, excessive session duration, and unauthorized attempts to access restricted internal resources.
3. Implement the source-defined work group: NFR-001: The solution shall be secure, scalable, reliable, and support both university-managed and approved personal devices.; SEC-001: All authentication events, access decisions, policy changes, and administrator actions must be logged.
4. In an authorized isolated test environment, execute TEST-001: Given: An authenticated user attempting remote access When: The user connects via the university VPN Expected: The connection is established with encrypted tunneling and validated credentials
5. In an authorized isolated test environment, execute TEST-002: Given: A user with elevated privileges attempting to access a restricted resource When: The user performs an action requiring MFA verification Expected: The system prompts for multi-factor authentication and grants access only after successful completion

6. In an authorized isolated test environment, execute TEST-003: Given: A guest user attempting to connect via the university VPN When: The system evaluates the user's authorization status Expected: The connection is denied with a clear message indicating unauthorized access
7. In an authorized isolated test environment, execute TEST-004: Given: An administrator with appropriate privileges When: The administrator initiates a session termination for a suspicious connection Expected: The system terminates the session and updates the audit log with a timestamp and reason
8. In an authorized isolated test environment, execute TEST-005: Given: A user attempting multiple failed logins from a new location When: The system monitors the login pattern and triggers an alert Expected: An automated security alert is generated and sent to the appropriate team
9. In an authorized isolated test environment, execute TEST-006: Given: A user with an approved personal device attempts to access campus resources from off-campus When: The system evaluates device compliance and authentication status Expected: The system shall permit access only if the device is university-approved or managed, and authenticated securely
10. In an authorized isolated test environment, execute TEST-007: Given: A user with a non-university-managed personal device attempts to connect via VPN When: The system performs device validation against approved device policies Expected: The system shall deny access and return a policy violation message
11. In an authorized isolated test environment, execute TEST-008: Given: A network administrator configures a new device policy for personal devices When: The policy is applied to the system configuration Expected: The system shall enforce the updated policy during subsequent access attempts
12. In an authorized isolated test environment, execute TEST-009: Given: An authenticated user performs a login or access decision When: The system processes the action Expected: A log entry is created with timestamp, user ID, action type, and outcome
13. In an authorized isolated test environment, execute TEST-010: Given: An administrator modifies a policy or revokes access When: The change is applied to the system Expected: A log entry is generated with timestamp, actor ID, policy name, and change details
14. In an authorized isolated test environment, execute TEST-011: Given: A user attempts an action that results in failure or success When: The action completes Expected: A log entry is created with outcome status and relevant context
15. Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
16. Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirement-linked implementation increment
2. Recorded test evidence
3. Reviewed feedback and next-iteration backlog

Entry criteria

1. Prior work, source scope and the reviewed test plan are available for iteration planning.

Exit criteria

1. A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.

Security checks

1. Review implementation and acceptance evidence against SEC-001: All authentication events, access decisions, policy changes, and administrator actions must be logged.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; NFR-001; SEC-001

Name: Release and handoff

Phase type: release

Activities

1. Review the exact requirements baseline, implementation links and verification evidence before a release decision.
2. Carry unresolved source decisions into handoff; generated text is not human approval.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirements and verification evidence package
2. Pending human release decision
3. Outstanding-decision record

Entry criteria

1. Verification evidence and unresolved decisions are available for human review.

Exit criteria

1. A human has recorded the release/handoff decision for the reviewed content.

Security checks

1. Review implementation and acceptance evidence against SEC-001: All authentication events, access decisions, policy changes, and administrator actions must be logged.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; NFR-001; SEC-001

Unknown factors: budget; delivery; failure_consequences; feedback; legacy; regulation; schedule; stability; uncertainty; verification

SDLC rankings and decision evidence

Policy: sequential-sdlc-1.1

Generation profile: sequential-sdlc-generation-2.0

Ranked candidates

Model: Agile

Score: 40

Reasons: No differentiating input evidence

Model: DevSecOps

Score: 40

Reasons: No differentiating input evidence

Model: Spiral

Score: 40

Reasons: No differentiating input evidence

Model: V-Model

Score: 40

Reasons: No differentiating input evidence

Model: Waterfall

Score: 40

Reasons: No differentiating input evidence

Model: Hybrid

Score: 40

Reasons: No differentiating input evidence

Model: Incremental

Score: 40

Reasons: No differentiating input evidence

Evidence register

Id: E-001

Source: description

Quote: Design a secure university VPN and remote-access management system that allows students, faculty, researchers, and administrative staff to securely access approved campus resources from off-campus locations.

Id: E-002

Source: description

Quote: The system should use centralized authentication, role-based access control, multi-factor authentication for privileged users, encrypted remote connections, session monitoring, and automatic security alerts.

Id: E-003

Source: description

Quote: Sensitive administrative and research systems should only be accessible to specifically authorized users, while guest users must not receive VPN access.

Id: E-004

Source: description

Quote: Network administrators should be able to create and revoke remote-access permissions, review active sessions, terminate suspicious sessions, manage VPN policies, and inspect audit logs.

Id: E-005

Source: description

Quote: The system should also detect repeated failed login attempts, unusual login locations, excessive session duration, and unauthorized attempts to access restricted internal resources.

Id: E-006

Source: description

Quote: All authentication events, access decisions, policy changes, and administrator actions must be logged.

Id: E-007

Source: description

Quote: The solution should be secure, scalable, reliable, and support both university-managed and approved personal devices.

Source references

kb-c1e0239af9dd46bd:1 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 1

Page or section: text chunk 2

Relevance score: 0.7633

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:8 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 8

Page or section: text chunk 9

Relevance score: 0.7275

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-0150d5352d0ec304:22 - 03_OWASP_CheatSheets/036_OWASP_Authentication_Cheat_Sheet.md

Source document id: kb-0150d5352d0ec304

Chunk index: 22

Page or section: text chunk 23

Relevance score: 0.7169

Sha256: b1d318a57f742d12b57bccda3153873653652c767038c5ca3d9d74350819f303

kb-0150d5352d0ec304:15 - 03_OWASP_CheatSheets/036_OWASP_Authentication_Cheat_Sheet.md

Source document id: kb-0150d5352d0ec304

Chunk index: 15

Page or section: text chunk 16

Relevance score: 0.7153

Sha256: b1d318a57f742d12b57bccda3153873653652c767038c5ca3d9d74350819f303

kb-ecd06e0e85ebe229:16 - 03_OWASP_CheatSheets/038_OWASP_Authorization_Cheat_Sheet.md

Source document id: kb-ecd06e0e85ebe229

Chunk index: 16

Page or section: text chunk 17

Relevance score: 0.7142

Sha256: e78a10eb9759a1fc7593ba1fc89a05b017e7e795ac16ad5496bdba86138d891f

Validation findings

Code: human_semantic_review_required

Severity: warning

Section: document

Message: Review completeness, actor permissions, contradictions and citation applicability; schema and link checks do not prove semantic correctness.

Human review history

None recorded.

AI drafting record

Requested model: cybersrs-qwen3-4b-ft:latest

Actual models used: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Section models

Introduction: cybersrs-qwen3-4b-ft:latest

Functional requirements: cybersrs-qwen3-4b-ft:latest

Non-functional requirements: qwen3:4b-instruct-2507-q4_K_M

Security, privacy and compliance: cybersrs-qwen3-4b-ft:latest

Compliance and policy applicability matrix: cybersrs-qwen3-4b-ft:latest

Architecture: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Threats: cybersrs-qwen3-4b-ft:latest

Risks: cybersrs-qwen3-4b-ft:latest

User stories and use cases: cybersrs-qwen3-4b-ft:latest

Verification and acceptance: cybersrs-qwen3-4b-ft:latest

Glossary: cybersrs-qwen3-4b-ft:latest

SDLC recommendation and workflow: cybersrs-qwen3-4b-ft:latest

Prompt version: sequential-1.28

Requirements traceability matrix

Requirement	Evidence / sources	Stories / tests / SDLC
FR-001	E-001	UC-001, TEST-001, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-002	E-002	TEST-002, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-003	E-003	TEST-003, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-004	E-004	TEST-004, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-005	E-005	TEST-005, Requirements and verification planning, Agile delivery cycle, Release and handoff
NFR-001	E-007	TEST-006, TEST-007, TEST-008, Requirements and verification planning, Agile delivery cycle, Release and handoff
SEC-001	E-006	TEST-009, TEST-010, TEST-011, Requirements and verification planning, Agile delivery cycle, Release and handoff

Model: cybersrs-qwen3-4b-ft:latest + qwen3:4b-instruct-2507-q4_K_M; prompt: sequential-1.28